Vivaan Bhandary

portfolio | 778-875-0090 | vivaanbhandary@gmail.com | linkedin | github

Education

University of British Columbia — Vancouver, BC
Bachelor of Science in Computer Science — *Sep. 2021 – May 2026*
GPA: 4.33, Dean's Honour List: 2022, 2023, 2024. Dean's Scholar 2026, Graduated with Distinction

Projects

Reloaded Dice - Video Game and Engine | *C++, Git, EnTT, Godot* — Jan 2026 – Apr 2026

- Directed a 6-person Agile team through 2-week sprints, integrating playtest feedback to refine core gameplay loops
- Architected a C++ game engine using the EnTT ECS framework for highly performant and scalable game logic
- Optimized core engine processes with the team by implementing spatial hashing and collision layers for efficient physics calculations, and resolved a major rendering bottleneck to reduce GPU overhead
- Designed and programmed a dynamic, synergistic upgrade system, managing complex entity interactions and state changes to scale gameplay mechanics
- Composed and integrated a dynamic music and sound system for the game, increasing player feedback for important actions and telegraphing gameplay elements

Graphics and Animation Projects | *JavaScript, GLSL, Python, speech_recognition* — Sept 2025 – Apr 2026

- Developed interactive 3D graphics engines with WebGL & custom GLSL shaders to render real-time environments
- Implemented multi-pass rendering for dynamic shadows and a multi-camera system to simulate real-time portals
- Engineered a voice-controlled facial animation system using state interpolation for smooth expression transitions
- Developed a custom animation pipeline with keyframe saving and inverse kinematics (IK) for fluid motion
- Wrote a physics-based cloth simulation engine with interactive collisions, particle forces, and custom tearing logic

Campus Information Query Engine | *TypeScript, Git, React, REST APIs* — Sept 2023 – Dec 2023

- Led Agile development of a TypeScript application, managing iterative sprints and weekly scrum meetings
- Parsed JSON and HTML campus data to create datasets for complex search and query operations
- Built secure RESTful API endpoints connecting the backend server with a React frontend
- Built a robust testing suite utilizing black-box, white-box, mutation, and unit testing

Data Analysis & Machine Learning Projects — Sept 2021 - Apr 2022

- Developed a K-Nearest Neighbors (KNN) regression model in R ($R^2 = 0.9$) to analyze and predict tennis player prize money based on ranking data
- Conducted hypothesis testing & generated bootstrap distributions using R to analyze regional demographics

Zone Chasing Game | *Java, Git, Java Swing* — Jan 2022 - Apr 2022

- Developed an object-oriented Java game utilizing the Swing library, featuring a dynamic GUI with custom gameplay animations, JSON-based data persistence, and a custom console logger to track object state changes

Experience

Junior Quality Assurance Engineer Co-op — Sept 2024 – Apr 2025
Trulioo — *Vancouver, BC*

- Progressed to Risk Team's **solo QA & QA Lead** within four months, managing end-to-end testing, deployments, and regression for critical risk monitoring systems
- **Optimized automated API test suites** for daily CI/CD pipelines, accelerating execution times by **4.5x**
- **Engineered comprehensive Postman collections** for **REST APIs** using dynamic variables and external data integrations to scale automated testing
- Developed and maintained mock API sandboxes to simulate third-party data sources and machine learning responses, enabling the dynamic testing of complex failure combinations and edge cases
- Led QA delivery for a high-priority data validation feature designed to intercept and quarantine false-positive data, managing test plans to ensure a zero-defect release under tight sprint deadlines
- Coordinated cross-functional testing strategies during a company-wide API standardization initiative, synchronizing automated test coverage across multiple distributed teams to ensure seamless integration

Artificial Intelligence Internship — July 2024 – Aug 2024
Safe Security — *Mumbai, India*

- Researched Retrieval-Augmented Generation (RAG) frameworks to optimize LLMs for accurate data retrieval
- Deployed a custom LLaMA architecture with RAG context to significantly improve context-aware AI responses

Technical Skills

Languages: C/C++, Java, Python, JavaScript, TypeScript, HTML, R, SQL, Erlang, PHP, Kotlin, GLSL
Frameworks & Libraries: React, Node.js, scikit-learn, Pandas, NumPy, Matplotlib, three.js, EnTT, NVIDIA warp
Software & Tools: Git, GitHub, Unity, Godot, Adobe Premiere Pro, VS Code, Visual Studio, PyCharm
Data Analysis & Visualization: Data Mining, Machine Learning, Data Visualization
Relevant Coursework: Video Game Programming, Computer Graphics & Animation, Machine Learning & Data Mining, Parallel Computation, Computer Networking, Computer Hardware & Operating Systems, Software Engineering & Construction, Algorithms and Data Structures, Data Science

Additional Certificates & Awards

1st Place at Trulioo's Fakeathon | *Company-wide Deepfake Detection* — Apr 2025
Wildest Idea Prize | *UBC Learning Analytics Hackathon* — Oct 2024
1st Place Prize | *UBC Game Design Competition* — Apr 2024